A New Fast Switching NBTI Characterization Method that Determines Subthreshold Slope Degradation during Stress

D. Brisbin*, P. Chaparala
National Semiconductor Corporation
2900 Semiconductor Drive, Santa Clara CA 95052-8090
*(408) 721-3642, Fax (408) 721-6454, douglas.brisbin@nsc.com

***Abstract*—For PMOSFET devices NBTI is a serious reliability concern. Because of recovery effects careful stress and measurement methods must be used to determine threshold voltage (V_T) degradation. These methods typically assume that mobility and subthreshold slope (SS) degradation are minimal. Recent papers have pointed out that this assumption may not be valid. This paper discusses a new fast switching NBTI measurement technique that alternates between two V_{GS} measurement conditions to determine the SS as a function of stress time. In this new method the SS is determined during stress to correctly compensate the V_T degradation for SS degradation. In addition, this paper presents SS and V_T NBTI degradation data from a 2 nm low and high nitrogen and a 6 nm DGO PMOS device to demonstrate the value of this new method.**

Keywords—NBTI, NBTI test methods, Negative Bias Temperature Instability, PMOSFET reliability

I. INTRODUCTION

NBTI is a serious reliability concern in state of the art PMOSFET devices [1-11]. The implementation of nitrided gate oxide to prevent boron penetration aggravates the NBTI issue [1-2,5]. Because of recovery effects [14-17] careful stress and measurement techniques (e.g. "On-The-Fly" and "fast V_{GS} switching") must be used to determine threshold voltage degradation [3, 12-14]. These methods assume that mobility and subthreshold slope (SS) degradation are minimal. Recent papers have pointed out that this assumption may be invalid [18-20]. This work discusses a unique NBTI measurement technique that alternates between two V_{GS} test conditions to determine the SS during stress to correctly compensate the V_T degradation for SS degradation. From these measurements the effect of SS degradation on V_T degradation can be accurately determined and results compared to the standard techniques.

II. EXPERIMENTAL DETAILS

The devices used in this study are a 1.2V PMOS with a W/L = 10/0.13 µm and an oxide thickness of 2 nm and a 3V DGO PMOS with a W/L = 10/0.3 µm and an oxide thickness of 6 nm. These devices are tested using the V_{GS} switching NBTI test method depicted in Fig. 1 [21]. In this technique the initial linear subthreshold I-V behavior of the device is obtained typically at V_{DS} = -0.05V and the constant current threshold voltage ($V_{T(ci)}$) and initial subthreshold slope SS(0) determined at a target drain current typically given by:

$$I_D = I_{D0} \times W/L \text{ with } I_{D0} = 50\text{-}100 \text{ nA}. \quad (1)$$

During the stress interval the V_{GS} stress is applied (with V_{DS} = 0V or -0.05V) for logarithmic half decade stress times (e.g. 1, 3, 10, 30, 100 etc.). After each stress time the magnitude of V_{GS} stress is reduced to a fixed constant V_{GS} @ test value equal to the initial $V_{T(ci)}$ and the drain current $I_D(t_{str})$ measured after a specified time delay (Δ). V_T degradation at stress time t_{str} is determined using:

$$\Delta V_T(t_{str}) = \log10(I_D(0) / I_D(t_{str})) / SS(0) \quad (2)$$

where $I_D(0)$ is the initial drain current at $V_{T(ci)}$. Eq. 2 is derived assuming that the subthreshold slope SS(0) is constant during stress as shown in Fig. 2. Stress termination occurs when a specified accumulative stress time is reached. At this time the final I-V behavior and post stress subthreshold slope $SS(t_{str})$ determined. Fig. 3a displays the stress-measurement time sequence of the proposed JEDEC standard technique [21]. Recently authors [18-20] have pointed out that assuming a constant SS(0) may not be valid and, therefore, the $\Delta V_T(t_{str})$ determined by the standard technique may be in error. Unfortunately, making multiple I_D measurements to determine the actual SS after any stress time interval is not accurate because of NBTI recovery effects.

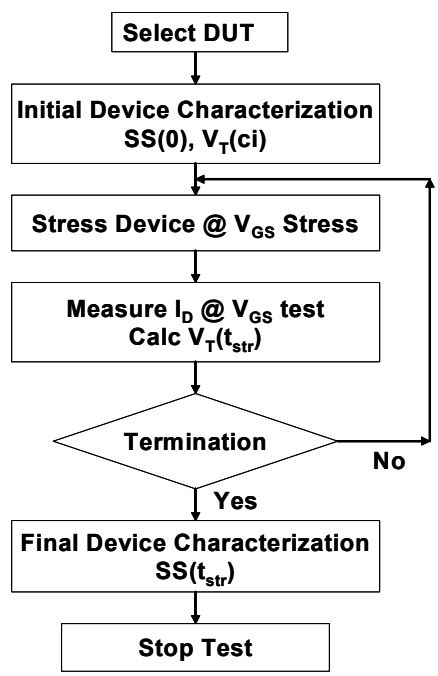


Figure 1. "Fast V_{GS} switching" test methodology.

Authorized licensed use limited to: National Institute of Technology- Delhi. Downloaded on April 14,2026 at 13:52:20 UTC from IEEE Xplore. Restrictions apply.

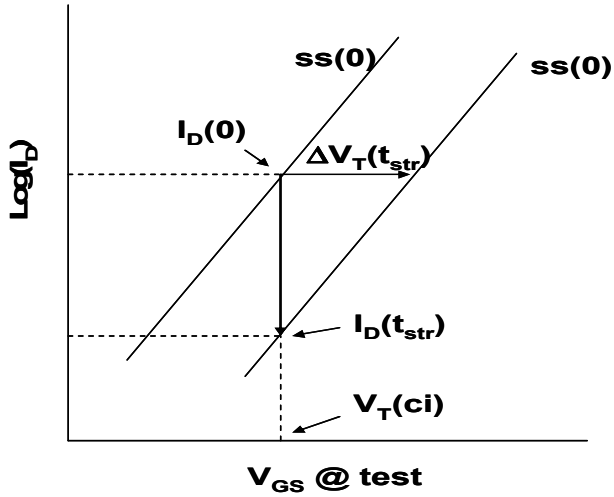


Figure 2. Method to extract ΔV_T from the measured $I_D(t_{str})$ and the initial SS(0).

This work differs from the standard technique in that two different V_{GS} test conditions are implemented. The first V_{GS} @ test is $V_{GS}1 = V_{T(ci)}$ while the second is $V_{GS}2 = V_{T(ci)} + \Delta V_{GS}$ (typically ΔV_{GS} = 20-50 mV). These two V_{GS} @ test conditions are applied at alternating stress intervals as seen in Fig. 3b. For most of the measurement results shown here, the source SMU was set for single line cycle average and a 3 ms delay (Δ) to allow for SMU settling time this resulted in a ~25 ms V_{GS} @ test pulse width. The fastest achievable test pulse for the instrumentation used here (no line cycle averaging) was ~4 ms which included a 1 ms settling delay (Δ). A fixed SMU measurement range is implemented during the stress measurement test sequence to prevent a momentary zero V_{GS} bias during SMU auto-ranging. After each stress time interval the $I_D(t_{str})$ is recorded. To validate the test methodology and measurement repeatability the proposed technique was used to obtain the SS in mV/decade vs. stress time for the V_{GS} stress condition of 0V (no stress). As seen in Fig. 4 the measurement repeatability is 1.2%.

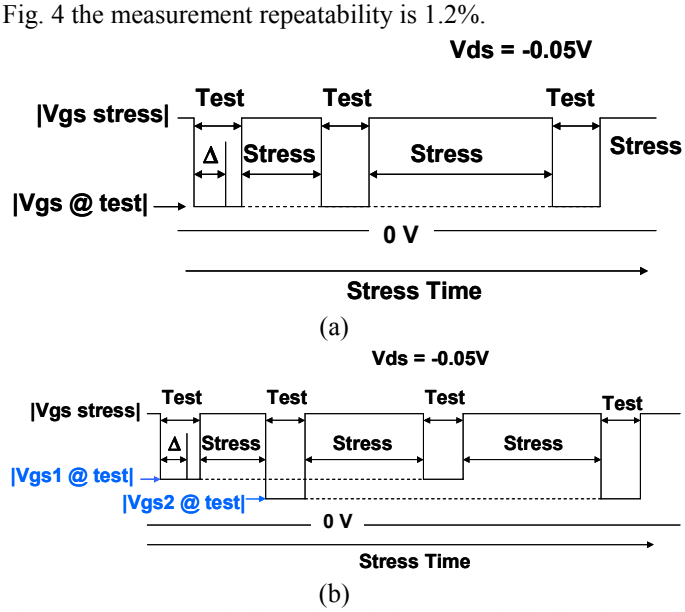


Figure 3. (a) Fast V_{GS} switching schematic (b) new method using alternate V_{GS} test voltages ($V_{GS}1$ and $V_{GS}2$).

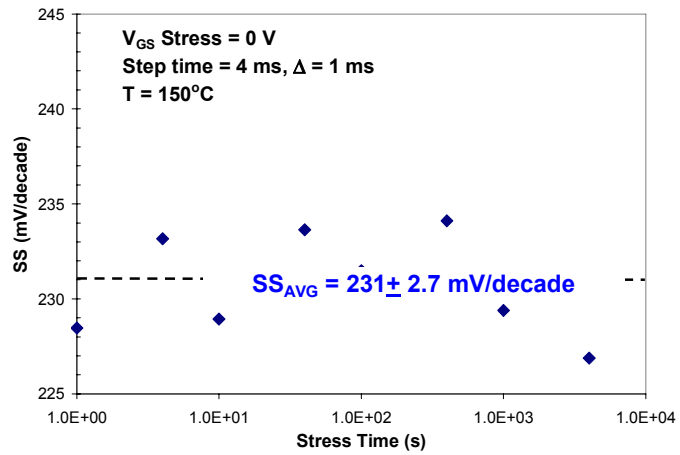


Figure 4. Measured SS vs. stress time for a V_{GS} stress condition of 0V for the PMOS(3V) device showing that the measurement repeatability is ±1.2% (1 sigma).

Fig. 5 displays typical I_D vs. stress time results for the two V_{GS} @ test conditions from the DGO PMOS(3V) devices where the number of stress intervals were increased (e.g. 1, 2, 3, 6, 10, 20, 30, 60, 100 etc.) to obtain improved line fit statistics. As expected each set of data has expected power law time dependence over several time decades. From fits of the data intermediate values of $\Delta I_D(t_{str})$ can be determined by interpolation at each measurement time. The $SS(t_{str})$ is determined using the following expression:

$$SS(t_{str}) = \log_{10}(I_D1(t_{str}) / I_D2(t_{str})) / \Delta V_{GS} \quad (3)$$

where $I_D1(t_{str})$ & $I_D2(t_{str})$ are the measured and interpolated I_D values at stress time t_{str}. The incremental V_T degradation is calculated using:

$$\Delta V_T(t_{str}) = \log_{10}(I_D(t_{strprev}) / I_D(t_{str})) / SS(t_{str}) \quad (4)$$

where $I_D(t_{strprev})$ is the measured I_D at the previous stress time interval.

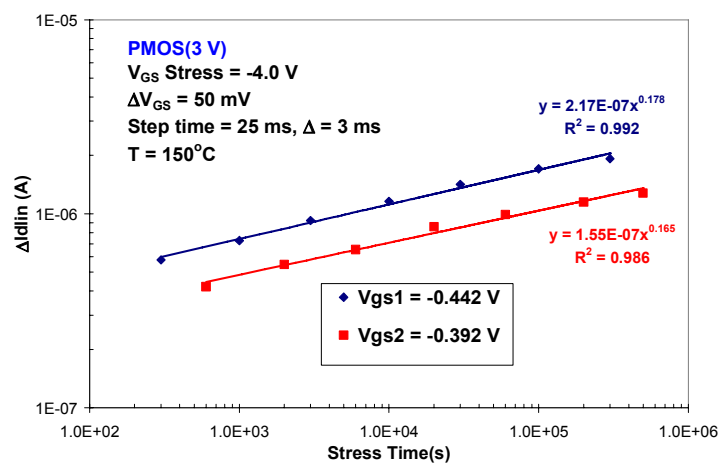


Figure 5. I_D degradation (ΔI_D) obtained using the two V_{GS} @ test method for the PMOS(3V) device.

Authorized licensed use limited to: National Institute of Technology- Delhi. Downloaded on April 14,2026 at 13:52:20 UTC from IEEE Xplore. Restrictions apply.

Fig. 6 displays calculated SS using (3) and (4) vs. stress time results from the $I_D(t_{str})$ data for the two devices shown in Fig. 5. The data shows that the SS for both devices is not constant but decreases (improving) with stress time. A decrease in SS with stress time is unexpected and seems to indicate that the turn-on characteristic of the device improves with stress. To understand this unexpected result the initial and post-stress SS data is plotted in Fig. 7 (note: the post-stress SS data is affected by NBTI recovery effects). Fig. 7 results show that indeed the SS decreases with stress time at the V_{GS} test condition and the magnitude of the decrease in SS is consistent with the results shown in Fig. 6. It is also apparent from Fig. 7 that SS is not constant with respect to V_{GS} as assumed by (2) but increases with increasing V_{GS}.

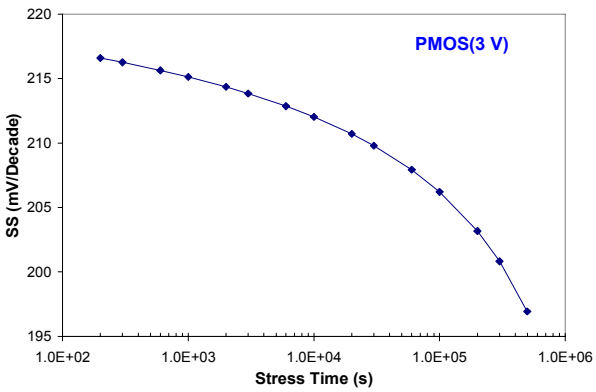


Figure 6. SS as a function of stress time for the PMOS(3V) device determined from Fig. 5 data.

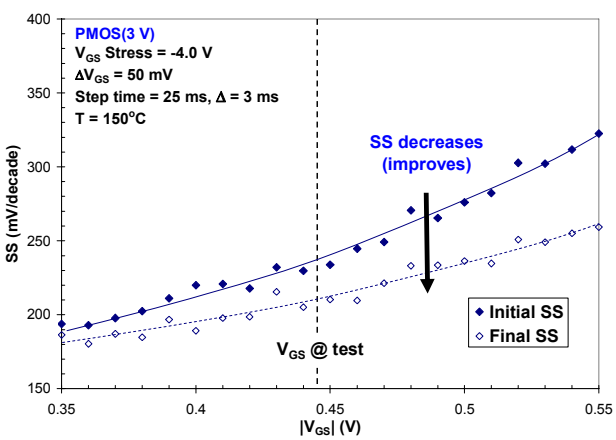


Figure 7. Initial and post-stress SS vs. V_{GS} for the PMOS(3V) device showing that SS does indeed improve (decrease) with stress time.

Fig. 8 explains the reason behind the SS improvement seen in Fig. 6. Fig. 8 displays the initial and post-stress I-V behavior with V_{DS} = -0.05V and the initial SS of the PMOS(3V) device stressed in Fig. 5. These results show that because of the NBTI induced increase in $|V_T|$ the I_D-V_{GS} curves shift to the right for constant I_D during stress effectively moving the fixed V_{GS} @ test point to a lower V_{GS} value and the steeper part of the SS curve. It is also apparent from Fig. 8 that the SS curves are not a constant until well below the target I_D (3.3 µA) of the device

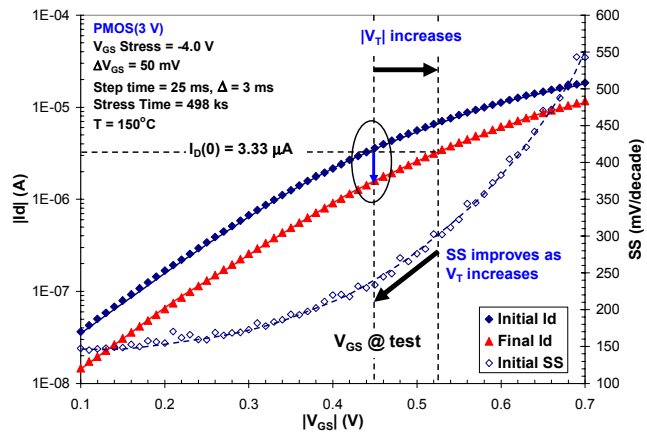


Figure 8. The cause of the SS improvement in PMOS(3V) device seen in Fig. 6 is a shift of the I_D-V_{GS} curve to the right during NBTI stress (increase in $|V_T|$) effectively moving the fixed V_{GS} @ test point to a lower V_{GS} value and the steeper part of the I_D-V_{GS} curve.

Figures 6-8 show that both the single and two V_{GS} @ test methods have a measurement flaw in that fixing the V_{GS} @ test typically results in the SS decreasing with stress time because $|V_T|$ increases during NBTI stress. One way to resolve this issue is to increment the V_{GS} @ test after each stress interval by the measured V_T degradation.

$$V_{GS} \text{ @ test} = V_{GS} \text{ @ test} + \Delta V_T(t_{str}) \quad (5)$$

As seen in Fig. 9 this keeps the V_{GS} @ test measurement point at the device $I_D = I_D(0)$ test point. Since the device I_D is returned to $I_D(0)$ after each stress interval the change in SS due to an increase $|V_T|$ during stress should be minimal.

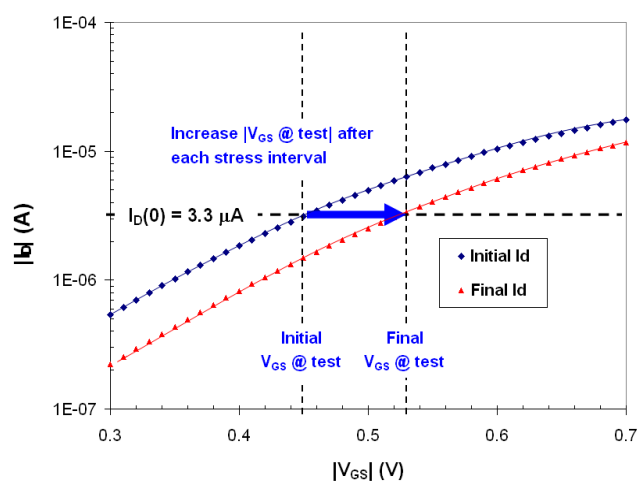


Figure 9. Incrementing the V_{GS} @ test after each stress interval using Eq. 2 effectively keeps the V_{GS} @ test measurement point at the device $I_D = I_D(0)$ test point.

INCREMENTAL TWO V_{GS} @ TEST METHOD

The two V_{GS} @ test SS data in Fig. 6 was obtained by a post fit of ΔI_D data where the two V_{GS} @ test values were fixed during stress. Unfortunately, this leads to a systematic

Authorized licensed use limited to: National Institute of Technology- Delhi. Downloaded on April 14,2026 at 13:52:20 UTC from IEEE Xplore. Restrictions apply.

measurement error as discussed above. To resolve this issue the V_{GS} @ test must be incremented during the stress interval.

Fig. 10 displays the incremental two V_{GS} @ test methodology. In this technique the $V_{GS}1$ @ test is incremented by the ΔV_T after each $V_{GS}1$ measurement interval. This effectively returns the device I_D to initial $I_D = I_D(0)$ (point A in Fig. 10). After the next stress interval the $I_D(tstr + 1)$ @ $V_{GS}2$ is determined where $V_{GS}2 = V_{GS}1 + \Delta V_{GS}$ and then finally after the next stress interval the device drain current $I_D(tstr + 2)$ @ $V_{GS}1$ is measured. The interpolated fit value of drain current $I_{DFIT}(tstr + 1)$ is determined using a logarithmic fit of the $I_D(tstr)$ and $I_D(tstr + 2)$ measurement points. Once I_{DFIT} is determined the $SS(t_{str})$ can be obtained using the following expression:

$$SS(t_{str} + 1) = \log_{10}(I_{DFIT}(t_{str} + 1) / I_D(t_{str} + 1)) / \Delta V_{GS} \qquad (6)$$

Since SS is a slowly varying function of stress time (see Fig. 6) $\Delta V_T(t_{str} + 2)$ can be determined using the following expression:

$$\Delta V_T(t_{str} + 2) = \log_{10}(I_D(0) / I_D(t_{str} + 2)) / SS(t_{str} + 1) \qquad (7)$$

Once $\Delta V_T(t_{str})$ is determined the increment in $V_{GS}1$ @ test can be obtained using (5).

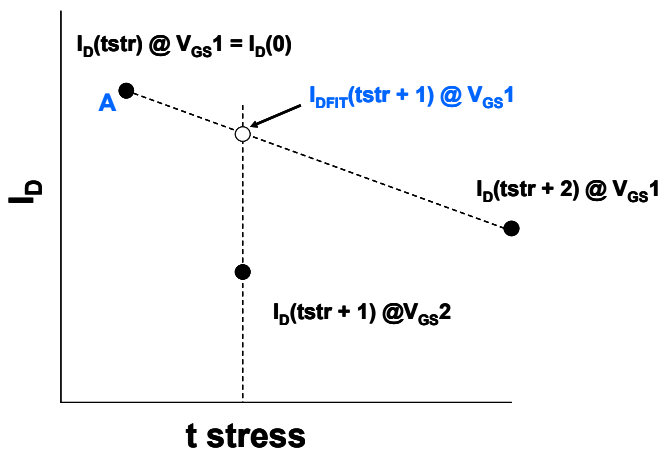
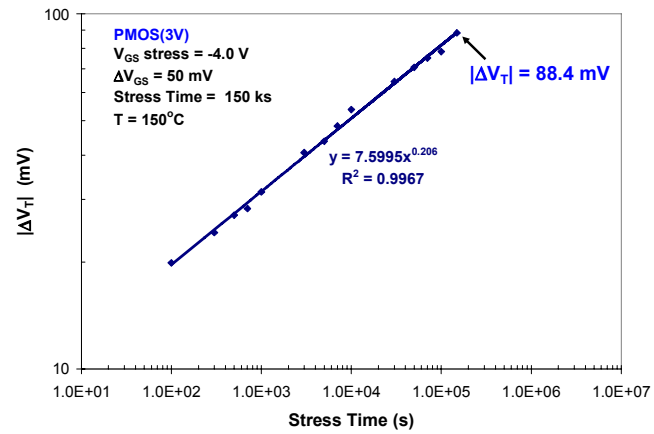
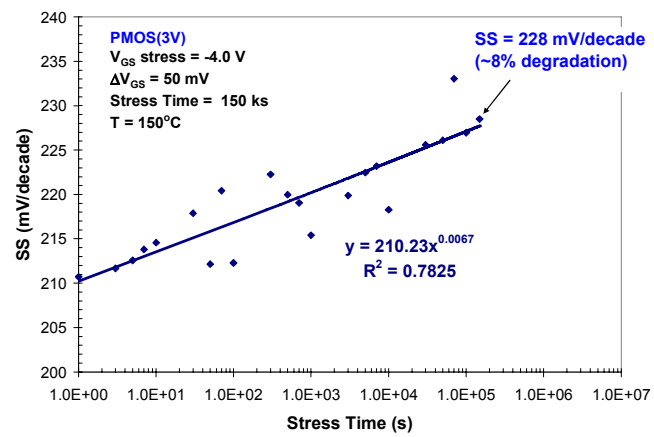


Figure 10. Schematic of the two V_{GS} incremental test method where the drain current at stress time (tstr+1) is obtained using interpolation and the SS(tstr) determined.

Fig. 11 displays NBTI V_T degradation and results from the PMOS(3V) device using this method where the total ΔV_T was 88.4 mV and the SS increase is 8% during the stress. Fig. 12 displays initial and post stress I_D and SS vs. V_{GS} results from the device stressed in Fig. 11. These results show that the total ΔV_T is 76 mV and SS increase is 7% in qualitative agreement with the value obtained in Fig. 11. It is important to point out as seen in Fig. 13 that assuming a constant SS = SS(0) during stress systematically underestimates the ΔV_T as determined by (7) by 8% (7 mV) for a stress time of 150 ks.



(a)



(b)

Figure 11. PMOS(3V) V_T degradation (a) and SS (b) where the V_{GS} @ test is incremented after each stress interval. As expected the SS increases (degrades) by ~8% while the SS measurement uncertainty is consistent with a 1.2% sigma (see Fig. 4).

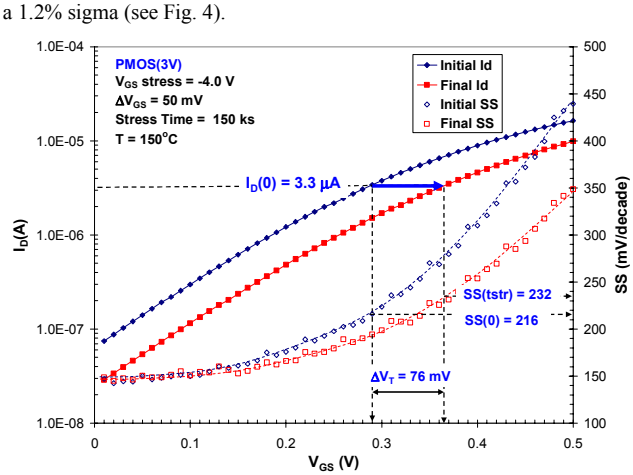


Figure 12. Pre- and post-stress I_D and SS results for the PMOS(3V) device stressed in Fig. 11. Results show that the ΔV_T and SS increase are in qualitative agreement with the value obtained in Fig. 11 with the SS degrading by 7% during stress.

Authorized licensed use limited to: National Institute of Technology- Delhi. Downloaded on April 14,2026 at 13:52:20 UTC from IEEE Xplore. Restrictions apply.

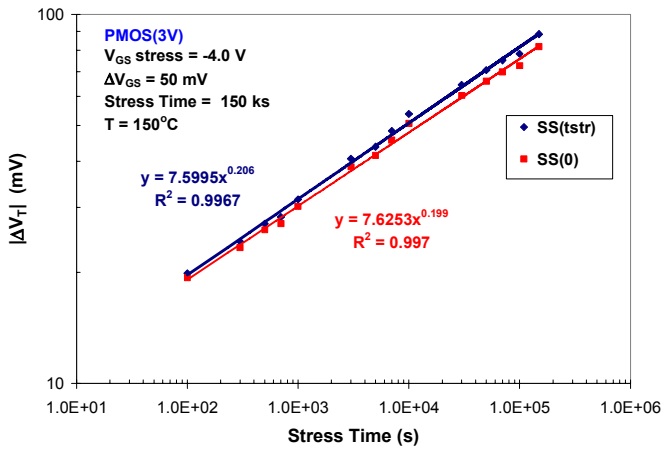
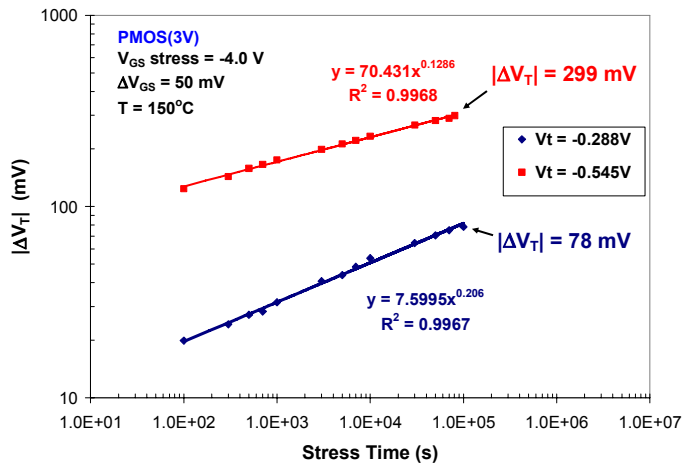


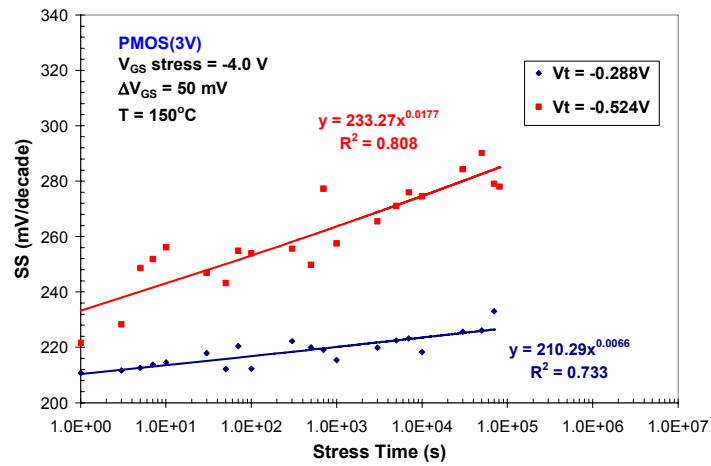
Figure 13. ΔV_T as determined using a constant SS = SS(0) and a stress time dependent SS(tstr). Assuming a constant SS systematically underestimates the ΔV_T during stress if SS degrades.

In order to understand the importance of SS degradation on NBTI measurement results two different types of devices were studied. In the first case, a 3V PMOS DGO device which exhibited substantial ΔV_T during stress and correlated to device $|V_T|$ [22] was studied. In the second case, a heavily nitrided 1.2V PMOS device (Tox = 2 nm) which was processed using decouple plasma nitridation (DPN) was examined.

Fig. 14 displays 3V PMOS device V_T and SS degradation results comparing a low and high $|V_T|$ device stressed using the same V_{GS}. As can be seen from Fig. 14 the high $|V_T|$ device had substantial V_T (299 mV) and a SS degradation (27%) compared to the low $|V_T|$ device where $|\Delta V_T|$ and ΔSS are 78 mV and 8%, respectively. In the case of the high $|V_T|$ device ignoring the SS degradation would result in significant underestimation (67 mV for 80 ks stress duration) of the NBTI induced ΔV_T. Pre- and post-stress sweep I_D and SS results (not shown) for the high $|V_T|$ device showed that the $|\Delta V_T|$ = 177 mV and the ΔSS = 19% in qualitative agreement but indicating considerable device recovery for the high $|\Delta V_T|$ device during the relatively slow V_{GS} sweep.



(a)

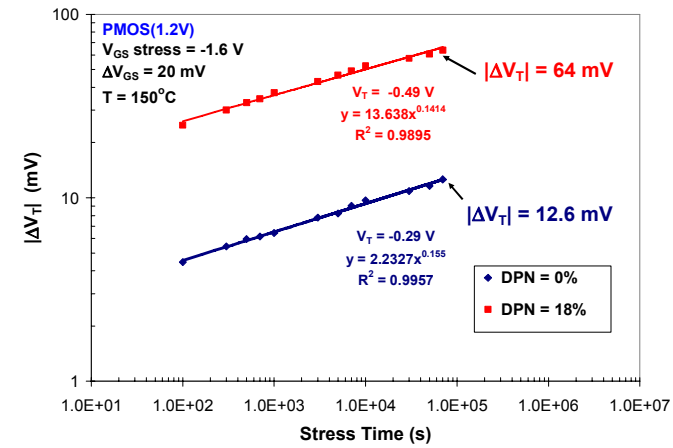


(b)

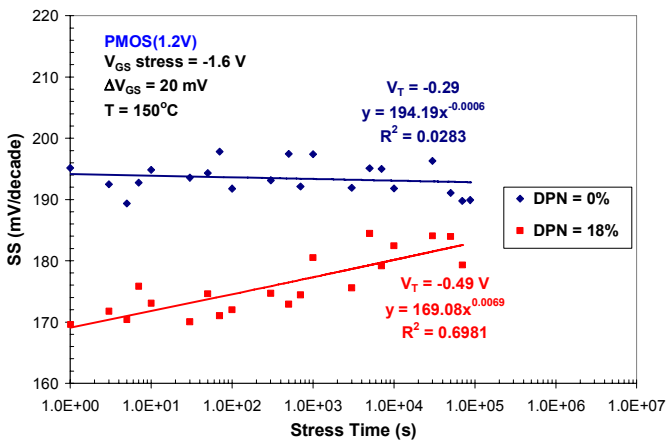
Figure 14. PMOS(3V) ΔV_T (a) and SS (b) for high and low V_T devices. For the high $|V_T|$ device the degradation was substantial with $|\Delta V_T|$ = 299 mV and ΔSS = 27% for a 80ks stress.

Fig. 15 compares 1.2V PMOS V_T and SS degradation results from a pure SiO_2 gate oxide film (0% DPN) to a heavily nitrided SiON film with 18 atomic % nitrogen (18% DPN) stressed at the same V_{GS}. As can be seen from Fig. 15a the $|\Delta V_T|$ is 64 mV and the ΔSS increase is 8% from the heavily nitrided gate oxide device compared to only 12.6 mV and 0% (within the measurement uncertainty) for the pure SiO_2 film. In the case of the 18% DPN device ignoring the SS degradation would result in an NBTI $|\Delta V_T|$ underestimation of ~ 5mV for 70 ks stress duration. Pre- and post-stress sweep I_D and SS results (not shown) for the 18% DPN device had a $|\Delta V_T|$ = 52 mV and a ΔSS = 5% while for 0% DPN device $|\Delta V_T|$ was 11 mV and the ΔSS increase 0.8% in qualitative agreement with the results displayed in Fig. 15.

Authorized licensed use limited to: National Institute of Technology- Delhi. Downloaded on April 14,2026 at 13:52:20 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)

Figure 15. PMOS(1.2V) ΔV_T (a) and SS (b) results for non-nitrided gate oxide device (0% DPN) and a heavily nitrided gate oxide device (18 atomic % nitrogen). For the heavily nitrided device the $|V_T|$ the degradation was substantial with a $|\Delta V_T|$ of 64 mV and ΔSS increase of 8% for a 70 ks stress.

III. Summary and Conclusions

A new fast switching NBTI measurement technique was introduced based on applying two alternating fixed V_{GS} test conditions to determine SS. NBTI results using this new method showed that, unexpectedly, SS decreases (improves) with stress time. It was determined that SS improvement was due to a measurement artifact that results from fixing V_{GS} @ test.

An improvement in the constant V_{GS} test methodology was proposed which consists of incrementing the V_{GS} @ test after each stress interval by the V_T degradation effectively keeping the device V_{GS} test point at the device V_T where $I_D = I_D(0)$. NBTI results from this improved method from a DGO PMOS(3V) device showed that as expected SS increases (degrades) with stress time and both ΔV_T and ΔSS results were in qualitative agreement with the post stress I_D-V_{GS} results. In addition, NBTI results from a heavily nitrided PMOS(1.2V) device and a high V_T PMOS(3V) device showed that SS degradation can be substantial leading to a significant underestimation of the NBTI V_T degradation during stress.

References

[1] A. Sakuma, D. Matsushita, K. Muraoka, and Y. Mitani, "Investigation of nitrogen originated NBTI mechanism in SiON with high nitrogen concentration", IRPS 2006, pp. 454-460.

[2] S.S. Tan, T.P Chen, J.M Soon, K.P. Loh, C. H. Ang, and L. Chan, "Nitrogen-enhanced negative bias temperature instability: An insightby experiment and first-principle calculations", Appl. Phys. Lett., vol.82, no. 12, pp.1881-1883, Mar. 2003.

[3] A. T. Krishnan, C. Chancellor, S. Chakravarthi, P.E. Nicollian, V. Reddy, A. Varghese, R.B. Khamankar, and S. Krishnan "Material dependence of hydrogen diffusion: implications for NBTI degradation", IEDM 2005, pp.688-691.

[4] D. Varghese, D. Saha, S. Mahapatra, K. Ahmed, F. Nouri, and M. Alam, "On the dispersive versus Arrhenius temperature activation of NBTI time evolution in plasma nitrided gate oxides: measurements, theory, and implications", IEDM 2005, pp. 684-687.

[5] Y. Mitani, "Influence of nitrogen on negative bias temperature instability in ultrathin SiON", IEEE Trans. Device Mater. Rel., Vol 8, No. 1, 2008, pp. 6-13.

[6] A.T. Krishnan, "NBTI: process, device and circuit", IRPS 2005 tutorial, pp.212.1-28.

[7] C. Schlünder, "Mixed signal circuit reliability – from device stress conditions to IC-failure", IRPS 2005 tutorial, pp.121.1-18.

[8] G. La Rosa, F. Guarin, S. Rauch, A. Acovic, J. Lukaitis, and E. Crabbe, "NBTI-channel hot carrier effects in PMOSFETs in advanced CMOS technologies", IRPS 1997, pp.282-286.

[9] D. Schroder and J. F. Babcock, "Negative bias temperature instability: road to cross in deep submicron silicon semiconductor manufacturing", J. Appl. Phys. Vol 94, pp.1-18, 2003.

[10] V. Reddy, A.T Krishnan, A. Marshall, J. Rodriguez, S. Natarajan, T. Rost, and S. Krishnan, "Impact of negative–bias temperature instability on digital circuit reliability", IRPS 2002, pp. 248-252.

[11] A.T. Krishnan, V. Reddy, S. Chakravarthi, J. Rodriguez, S. John, and S. Krishnan, "NBTI impact on transistor and circuit: models, mechanisms and scaling effects", IEDM 2003, pp.349-352.

[12] T. Grasser, P. Wagner, P. Hehenberger, W. Goes, and B. Kaczer, "A rigorous study of measurement techniques for negative bias temperature instability", IEEE Trans. Device Mater. Rel., Vol 8, No. 3, pp. 526-535, 2008.

[13] M. Denais, C. Parthasarathy, G. Ribes, Y. Rey-Tauriac, N. Revil, A. Bravaix, V. Huard, and F. Perrier, "On-the-fly characterization of NBTI in ultra-thin gate oxide PMOSFETS", IEDM 2004, pp. 109-112.

[14] H. Reisinger, O. Blank, W. Heinrigs, A. Muhlhoff, W. Gustin, and C. Schlunder, "Analysis of NBTI degradation and recovery behavior based on ultra fast V_T measurements", IRPS 2006, pp. 448-453.

[15] C. R. Parthasarathy, M. Denais, V. Huard, G. Ribes, E. Vincent, and A. Bravaix, "New insights into recovery characteristics post NBTI stress", IRPS 2006, pp. 471-477.

[16] S. Rangan, N. Mielke, and E.C.C. Yeh, "Universal recovery behavior of negative bias temperature instability", IEDM 2003, pp. 341-344.

[17] T. Grasser, W. Gos, V. Sverdlov, and B. Kaczer, "The universality of NBTI relaxation and its implication for modeling and characterization", IRPS 2007, pp. 268-280.

[18] C. Schlünder, M. Hoffmann, R.-P Vollertsen, G. Schindler, W. Heinrigs, W. Gustin, and H. Reisinger, "A novel multi-point NBTI characterization methodology using smart intermediate stress," IRPS 2008, pp. 79-86.

[19] A. Islam, V.D. Maheta, H. Das, S. Mahapatra, and M.A. Alam, "Mobility degradation due to interface traps in plasma oxynitride PMOS devices", IRPS 2008, pp. 87-96.

[20] D. Brisbin and P. Chaparala "The effect of the subthreshold slope degradation on NBTI device characterization", IIRW 2008, pp. 96-99.

[21] JEDEC 14.2.2, "A procedure for wafer level DC characterization of P-MOSFET Negative Bias Temperature Instabilities" to be published.

[22] D. Brisbin, J. Yang, S. Bahl, and C. Parker, "PMOS NBTI degradation due to halo implant channeling", IRPS 2008, pp. 61-65.

Authorized licensed use limited to: National Institute of Technology- Delhi. Downloaded on April 14,2026 at 13:52:20 UTC from IEEE Xplore. Restrictions apply.